

Regional Basins and the Social Fabric: Fragility, National Regimes, and the Regulation of Development in a Heterogeneous-Agent Economy

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Abstract

A companion empirical paper shows that European regional development is a nationally-structured system of basins of differing depth, revealed by a common shock: struggling regions retain their position about ninety per cent of the time while high-performing regions retain theirs only fifty-seven per cent, with the largest single flow a drift of high-performing regions down to the stable core. This paper asks whether a disciplined heterogeneous-agent model of participation in social life reproduces that asymmetry from independent calibration, prices it, and identifies what regulates it. Three results. First, a discipline result: at the calibration that fits a real participation gradient there are no multiple equilibria, so the asymmetry cannot rest on a tipping point; basin depth is instead graded social amplification, the local multiplier $1/(1 - s)$ where s is the slope of the participation map at its fixed point. Calibrated to the four tiers, the equilibrium collapse of the social fabric under a common agency shock rises monotonically with development, mirroring the measured retention with the same shape, flat at the bottom and a cliff at the top. Second, a population of regions hit by one shock reproduces the empirical transition matrix, including the high-performing drift, and when regions are nested in national regimes the downgrades cluster in high-development nations, the nationally-structured transition the data show. Third, the shock is priced at minus sixteen hundred pounds per person per year in the high-performing tier against minus nine in the struggling tier, and two policy levers separate: raising agency lifts the whole landscape but moves regions into the shock-sensitive zone, while a participation tax credit holds the coupling in place, collapsing fragility where it is highest. Prosperity is fragile because it is a coordination achievement; poverty is a floor with little fabric to lose.

*Draft. Third in a series with “Wellbeing and the Social Fabric in a Heterogeneous-Agent Economy” and “Agency and the Social Fabric”; the three papers share one Julia engine. Every number and figure here is reproduced by the scripts in `paper3/` in the public repository, which read the four ecosystem tier centroids and transition matrix from the companion empirical paper.

1 Introduction

This is the modelling paper in a three-paper arc on inclusive regional development. The first proposes a coupled three-pillar framework of Place-based Conditions, Human and Social Capital, and Economic Activity. The second shows empirically that, across European regions, the framework describes a nationally-structured system with regional basins of differing depth, and that a common shock reveals the structure: a four-tier development typology (struggling, catching-up, stable core, high-performing) that is stable in calm times but whose top tier is a shallow perch and whose bottom tier is a deep trap. The single most striking fact is an asymmetry in retention under the shock, ninety-two per cent for the struggling tier against fifty-seven for the high-performing, with downward moves dominating and the largest flow a drift from high-performing to stable core.

The question here is whether a standard incomplete-markets model, extended with a social-participation margin and calibrated independently to those tiers, generates that asymmetry as an output, prices it in wellbeing, and says what regulates it. The discipline of the exercise is that the basin depths must not be tuned in: they have to fall out of a calibration anchored to facts the model was not aimed at.

The model is the participation economy of the companion agency paper. Households split time between paid work and a discrete choice to take part in associative life, where the value of joining rises with how many others join, a Brock-Durlauf complementarity [Brock and Durlauf, 2001]. That paper established two things this one uses: the complementarity is a coordination device that, on the discrete margin, can support multiple equilibria; and, at the calibration that fits the French participation gradient, the economy sits at the boundary of the multiplicity region rather than inside it. The contribution here is to take that boundary economy to the regional cross-section, to show that the empirical basin asymmetry is reproduced not by tipping but by graded amplification, and to make the national overlay an explicit two-level object.

2 Model

2.1 Regions as participation economies

A region is a population of households facing the problem of the companion paper. The household state is assets $a \geq 0$ and idiosyncratic productivity z following a two-state Markov chain Π (Rouwenhorst discretisation, $\mathbb{E}[z] = 1$), with a riskless asset at exogenous gross rate R . Each period the household chooses consumption c , savings $a' \geq 0$, market effort e , and a participation indicator $d \in \{0, 1\}$; participating commits a minimum time lump $\bar{q} = 0.10$, so $e + \bar{q}d \leq 1$. There are two permanent education groups $g \in \{l, h\}$ with agency α_g (the share of effort that becomes income) and belonging taste B_g ; within each group, households differ in a permanent taste m , lognormal with unit mean and log standard deviation σ_m . The budget is

$$c + a' = R a + (1 + \tau) \alpha_g e z - T_x, \quad a' \geq 0, \quad (1)$$

with τ a make-work-pay subsidy and T_x a balancing lump-sum tax. The material block of utility is the separable $U^c = \Gamma\left(\frac{e^{1-\gamma}}{1-\gamma} - \phi \frac{e^{1+\psi}}{1+\psi}\right)$, the wealth-effect channel retained as in the companion papers. The payoff to participating, per unit of committed lump, is

$$\kappa \Lambda B_g m (\omega + (1 - \omega) r), \quad (2)$$

where r is the aggregate participation rate, κ the interaction strength, Λ the cohesion weight, ω the private warm-glow share, and $(1 - \omega)$ the complementarity share. The social fabric is $Q = \bar{q} r$. Decision utility moves behaviour through (2); experienced wellbeing is priced separately on the $\Lambda B_g Q$ dashboard, as in the companion papers.

2.2 From development position to agency and belonging

The empirical typology is, to a very good approximation, one-dimensional: a single general factor reproduces the four tiers almost exactly, and the three pillars correlate between 0.78 and 0.89. The four tier centroids are therefore four positions on one development axis. We map a tier's level L (the mean of its pillar centroids) to household ingredients by an affine rule anchored to France, which sits in the stable-core tier and is the one point the model is already calibrated to:

$$\alpha_l(L) = \alpha_l^{FR} + s_\alpha (L - L^{\text{core}}), \quad \alpha_h(L) = \alpha_h^{FR}, \quad (3)$$

$$B_l(L) = B_l^{FR} + s_B (L - L^{\text{core}}), \quad B_h(L) = B_h^{FR}. \quad (4)$$

The asymmetry between floor and ceiling is not a modelling convenience: the cross-country calibration of the shared engine shows that development acts almost entirely on the low-education agency floor (α_l ranges from 0.55 to 0.77 across countries) while the high-education ceiling barely moves (0.88 to 0.93). So the within-region agency gap widens as development falls, endogenously, and no separate regional inequality series is needed; the inequality channel is inside $\alpha_l(L)$. The slopes s_α and s_B are anchored to that cross-country range and swept for robustness.

2.3 Nesting regions in national regimes

The companion empirical paper finds that most of the variation in regional development is between countries. We encode this as a two-level object. Each country is a regime with a mean development level L_n , which sets its agency floor and gap, the “load” that fixes the basin landscape. Each region's level is $L = L_n + \xi$, with ξ a within-country texture deviation. The participation equilibrium is then a nested fixed point,

$$G(r; \theta_n) = r, \quad (5)$$

where the national parameter θ_n shifts the map a region's rate solves, directly analogous to the per-group aggregate of the homophily channel in the first companion paper. The national layer carries the overlay; the regional layer is the texture within it.

Table 1: Fragility by tier at the calibrated point ($\omega = 0.30$).

tier	equilibrium rate	map slope s	multiplier $1/(1 - s)$	fabric fall under shock
struggling	0.248	0.661	2.95	0.0003
catching-up	0.292	0.679	3.11	0.0066
stable core	0.345	0.705	3.39	0.0286
high-performing	0.408	0.755	4.08	0.0497

2.4 Basin depth as social amplification

At a region’s stable equilibrium the aggregate map G has slope s . The local social multiplier is $1/(1 - s)$, and $1 - s$ is a continuous distance-to-tipping: as $s \rightarrow 1$ the map becomes tangent to the diagonal and the region reaches a fold. This object is well defined whether or not a second equilibrium exists, it reduces to the strict distance-to-tipping in the limit, and it is the same multiplier that the companion paper measures as the policy amplification ratio. Basin depth, the resistance of a region’s fabric to a common shock, is read off as the equilibrium fall in participation when agency is lowered uniformly; the multiplier governs how much a given direct hit is amplified.

3 Calibration

The social technology is treated as common to all regions: κ , σ_m and ω are human, not national, and only the agency and belonging endowments differ by tier. We fix $\omega = 0.30$ (swept over $\{0.15, 0.30, 0.50\}$) and calibrate (κ, σ_m) to the French group participation rates, low-education 0.25 and high-education 0.45, at the stable-core endowment. The fit is $\kappa^* = 10.25$, $\sigma_m^* = 0.65$, delivering group rates 0.258 and 0.433 at an aggregate of 0.345. At this point the region has a single equilibrium and no fold: France sits on the boundary, as the companion paper found. The preference and process parameters are inherited at their sourced values ($\gamma = 2$, $\psi = 2$, $\beta = 0.96$, $R = 1.02$, $\phi = 14$, the agency and belonging anchors from the OECD Better Life indicators by education).

4 The fragility gradient

Mapped to the four tiers, the equilibrium participation rates are strictly ordered (0.248, 0.292, 0.345, 0.408) and every tier has a single equilibrium. There is no fold anywhere on the development ladder at the honest calibration, and this survives all swept slopes. This is the discipline result: the basin asymmetry does not come from multiple equilibria. It comes from amplification. Table 1 reports, per tier, the equilibrium fall in the social fabric under a common five-point agency shock, together with the map slope and the multiplier.

The fabric fall rises monotonically with development, and the prosperous tier’s fabric falls about one hundred and sixty-five times more than the struggling tier’s under the same shock. This mirrors the empirical retention (Figure 1), and it mirrors it in shape, not merely in rank: retention is flat across the bottom two tiers (both ninety-two per cent), barely down in the core, then a cliff to the

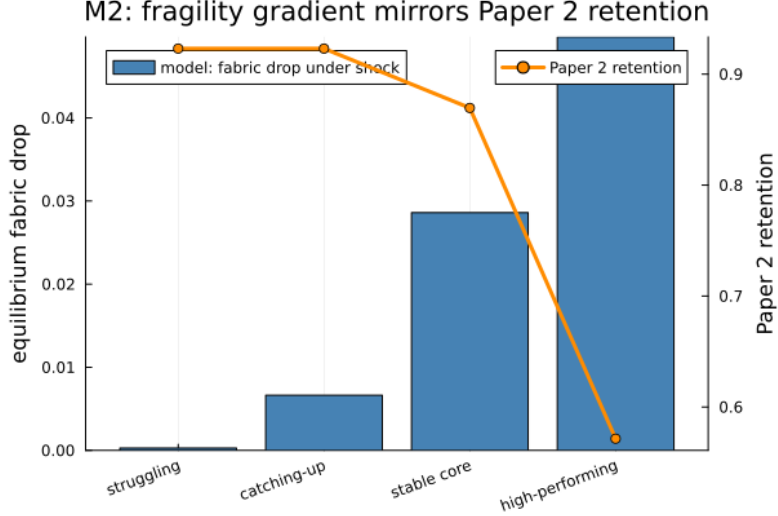


Figure 1: Model fabric fall under a common shock (bars) against the measured retention (line), by tier. The flat-bottom-and-cliff shape, not just the rank order, matches.

top, and the model’s fabric fall has the same flat-bottom-and-cliff profile. The flat bottom is a floor: a thin-fabric region’s households are too far from the participation margin for a shock to move them. The cliff is amplification biting where participation is dense. The honest decomposition is that the multiplier is roughly constant at about three across tiers, so the gradient is carried by the direct effect, which is far larger at the top because more households sit near the margin and there is more fabric to lose. The direction holds across the swept ω , and the seed-averaged result is stable. One refinement: at small shocks the most reactive tier is the core, which sits on the boundary; only at the larger, pandemic-scale shock does the high-performing tier become the most fragile. Small shocks bite the boundary, large shocks bite the prosperous.

5 The common shock and the transition matrix

We build a population of 294 regions with the empirical tier counts, scattered within tier, and apply one common agency shock. A region moves down a tier when the resulting fabric collapse, mapped to the development level, carries it across the band below. The single free scale of that mapping is set so the total number of downgrades matches the data, about fifty of 294; the distribution of downgrades across tiers is then a prediction. Table 2 compares the model and empirical matrices.

Forty-seven of the model’s forty-nine downgrades are high-performing regions falling to the stable core, against forty-one in the data; overall persistence is eighty-three per cent against seventy-nine. The dominant empirical flow is reproduced as model output. Across twenty-five random populations the high-performing retention is 0.51 (standard deviation 0.014) and ninety-seven per cent of downgrades are high-performing, so the concentration is structural. The model’s lower tiers are cleaner than the data because it contains only the common shock; adding an idiosyncratic component, calibrated to overall persistence, reproduces the small lower-tier churn while holding

Table 2: Model (common shock) and empirical transition matrices; rows pre-shock, columns post-shock.

	struggling	catching	core	high	retention
<i>Model</i>					
struggling	39	0	0	0	1.00
catching-up	0	65	0	0	1.00
stable core	0	2	90	0	0.98
high-performing	0	0	47	51	0.52
<i>Empirical</i>					
struggling	36	0	0	3	0.92
catching-up	2	60	3	0	0.92
stable core	0	6	80	6	0.87
high-performing	0	1	41	56	0.57

the high-performing collapse. The model slightly over-predicts that collapse (0.52 against 0.57), which is the right direction given the data’s own caveat that the figure is inflated by post-Brexit attenuation in the UK regions.

6 The national overlay

Making the two-level structure explicit, we place regions in national regimes with the between-country share of the development level set to a high value consistent with the empirical finding (the exact share is an empirical quantity to be recomputed). The model implication is what matters. The between-nation share of fragility is 0.79 against 0.87 for the level, so the overlay carries through to basin depth nearly one-for-one, with the nonlinearity of the multiplier adding modest within-nation spread. Under a common shock the downgrade rate correlates with the national mean level at 0.82, running from three per cent in the bottom development quartile of nations to forty-five per cent in the top. The nationally-structured, asymmetric transition is therefore reproduced at the two-level scale: national regimes set the load, fragility accelerates with it, and the shock concentrates downgrades in the prosperous nations. What is left out is explicit: the strong spatial autocorrelation in the data is a cross-sectional dependence a partial-equilibrium model does not carry, and the empirical paper’s own finding that the spatial graph adds essentially nothing beyond the indicators is the warrant for omitting it here. It is signposted for a general-equilibrium and spatial extension.

7 Pricing and policy as basin regulation

Using the wellbeing bridge of the companion work, the social-support coefficient of the World Happiness Report and the Treasury value of thirteen thousand pounds per life-satisfaction point, the common shock is priced at minus nine pounds per person per year in the struggling tier against minus sixteen hundred and twenty-three in the high-performing tier (Figure 2). The shock is

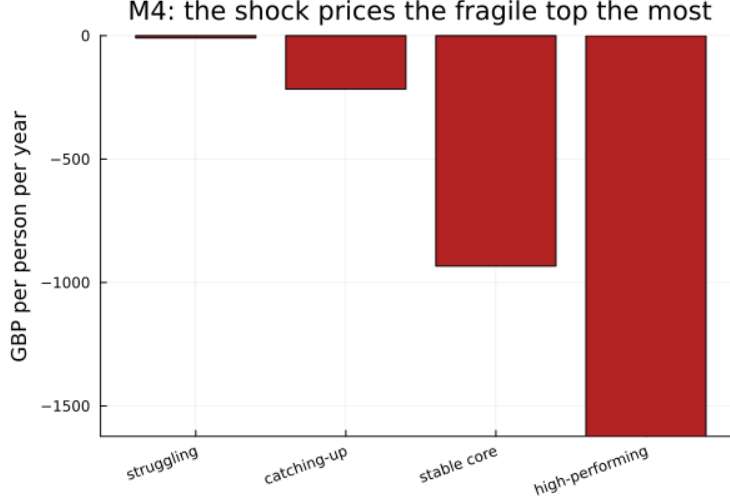


Figure 2: Wellbeing price of the common shock by tier, pounds per person per year.

regressive in reverse on the fabric dimension: the prosperous, fragile top pays the most because it has the most coordinated fabric to lose. This is the price of the shallow perch.

Two levers regulate the basin, and they are different. Raising the low-education agency to the population mean (empowerment) lifts participation in every tier, reproducing the companion paper's thirty-seven to fifty-one per cent rise, but it moves regions up into the coordination-sensitive zone, so their fabric response to a shock grows: it changes the load, lifting the whole landscape, but does not by itself hold the coupling. A participation tax credit, by contrast, rewards joining directly, so each region depends far less on others. It lifts participation toward universal and collapses fragility: the high-performing tier's multiplier falls from 4.08 to about 1.04 and its fabric response to a shock falls by more than eighty per cent. The credit is the lever that holds the coupling, and it does most where fragility is highest. The make-work-pay subsidy is the opposite, a load shock: financed, the substitution effect dominates the income effect and participation falls in every tier, by most in the high-performing tier (-0.21) and least in the struggling tier (-0.12), so the load shock, like the common shock, bites the fragile top hardest. The mechanism is the thirty-seven to eight per cent collapse of the companion paper, milder here because this calibration sits a little further from the fold. Three pro-prosperity policies, three different effects on the basin: raise it, hold it, or knock it over. A single-index evaluation sees none of the difference.

8 Discussion

The claim is bounded, and the bound is the point. This is not a growth model: with the interest rate exogenous it is a model of a stationary distribution and its comparative statics, not of accumulation or structural transformation, and it does not explain how a region grows out of a trap over time. What it provides is a microfoundation for development on the wellbeing-and-resilience dimension: why regions at different development levels carry differently fragile social fabric, what that fragility

is worth, and which levers reshape it. It complements, rather than competes with, the agglomeration and geography-of-discontent account of how regions arrived where they are [Rodríguez-Pose, 2018]. The agency parameter is the natural structural home for a capability reading of development, the wealth distribution carries the material side, and the participation complementarity supplies the coordination mechanism that makes prosperity an achievement and therefore fragile. The limitations are those of the model class and are stated rather than apologised for: partial equilibrium, no spatial channel, a wellbeing aggregator whose non-substitutability is the substantive and contestable assumption, and a nesting whose between-country share is taken from the empirical paper rather than derived.

9 Conclusion

A disciplined participation economy, calibrated independently to four regional development tiers, reproduces the basin-depth asymmetry the empirical paper measured, including the concentration of downward moves in the prosperous tier and their clustering in prosperous nations. It does so not through a tipping point, which the honest calibration rules out, but through graded social amplification, which makes the result robust to the elasticities that sank the earlier tipping claim. It prices the asymmetry and separates the policy levers that regulate it. The substantive reading is simple: prosperity is fragile because it is a coordination achievement that many people sustain together, and a common shock that makes a few step back makes many step back; poverty is a floor with little fabric to lose. Where the social fabric is multidimensional, that asymmetry is invisible to the ledgers that development policy usually keeps.

References

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